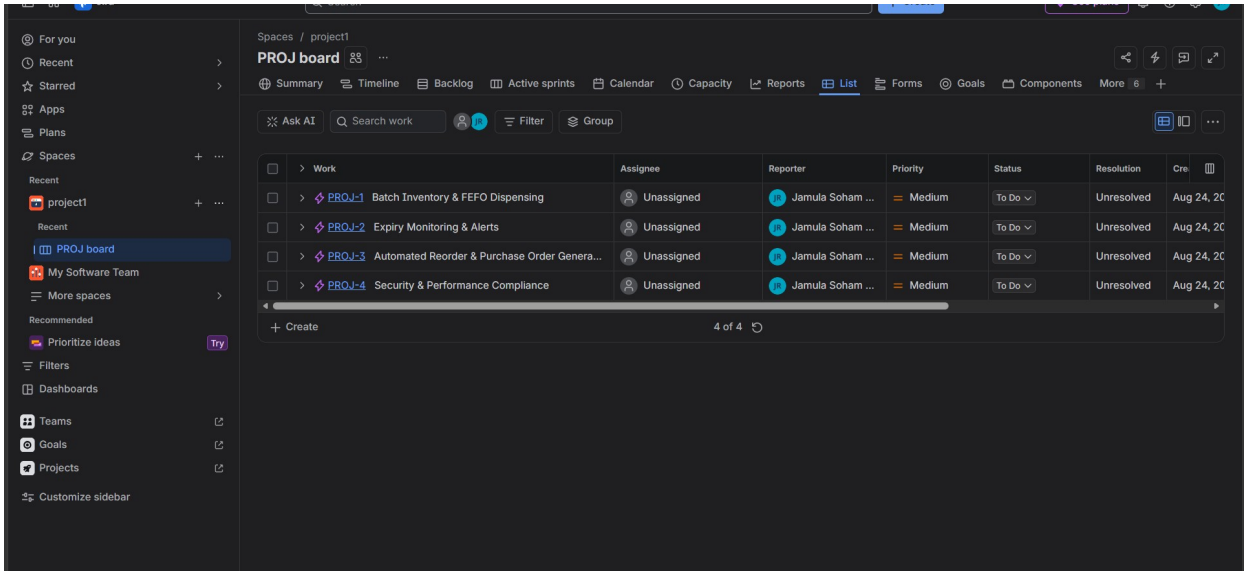
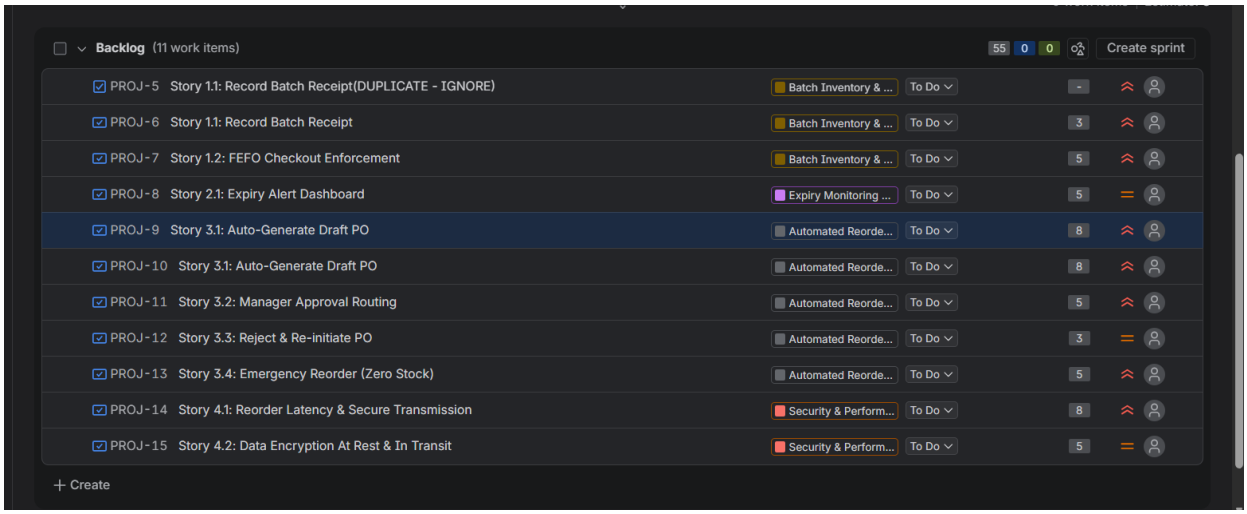
Lab 2

Problem Statement #15 — Pharmacy Expiry & Re-order Dispatch Engine

Jamula Soham Rao | PES2UG24CS202

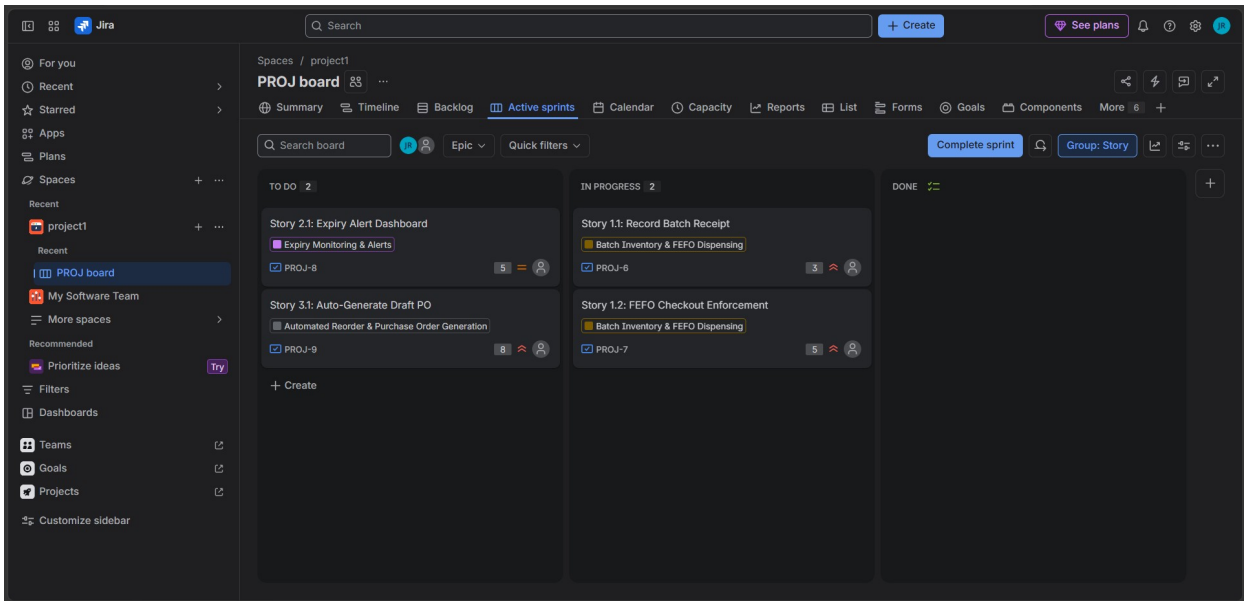
1. Backlog: Epics and User Stories

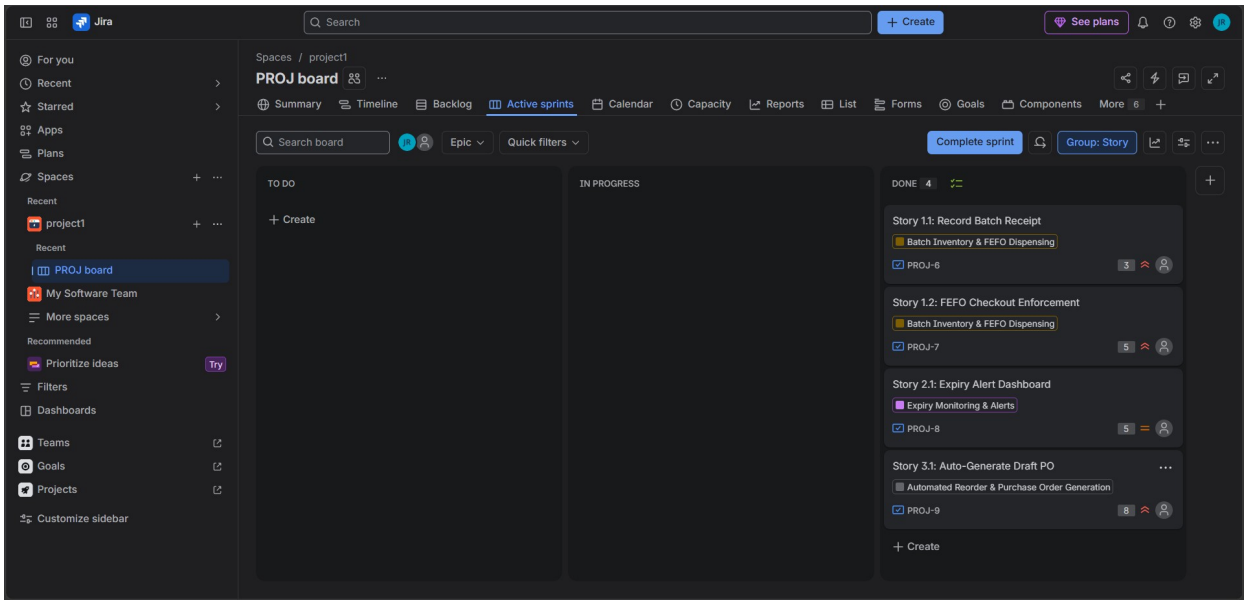




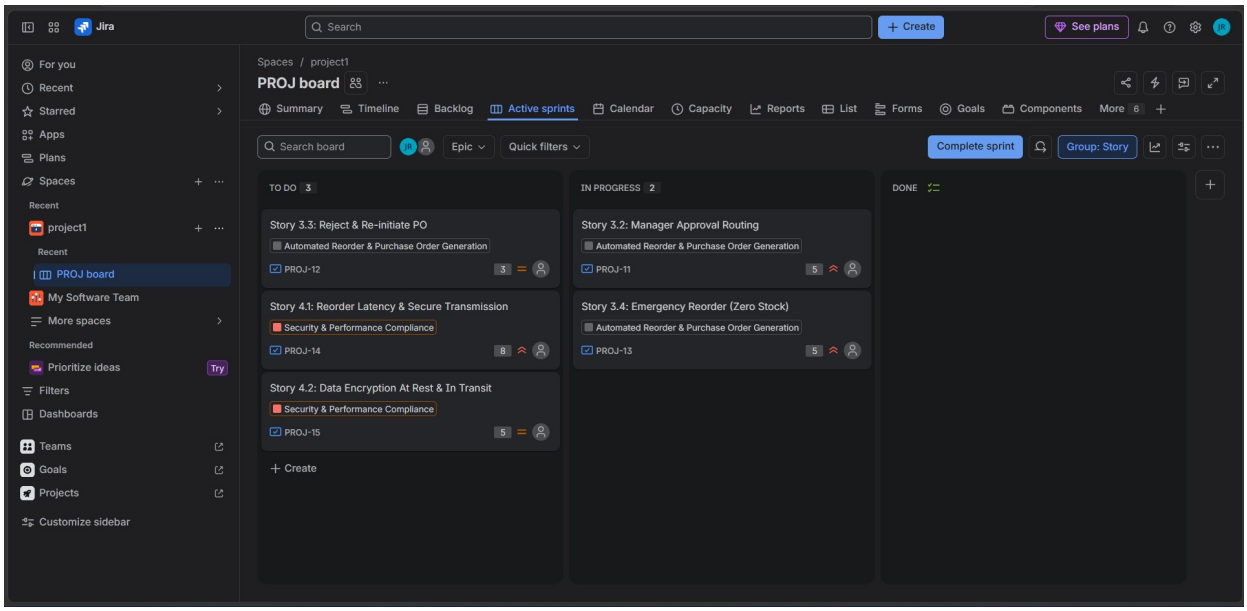
2. Sprint Execution

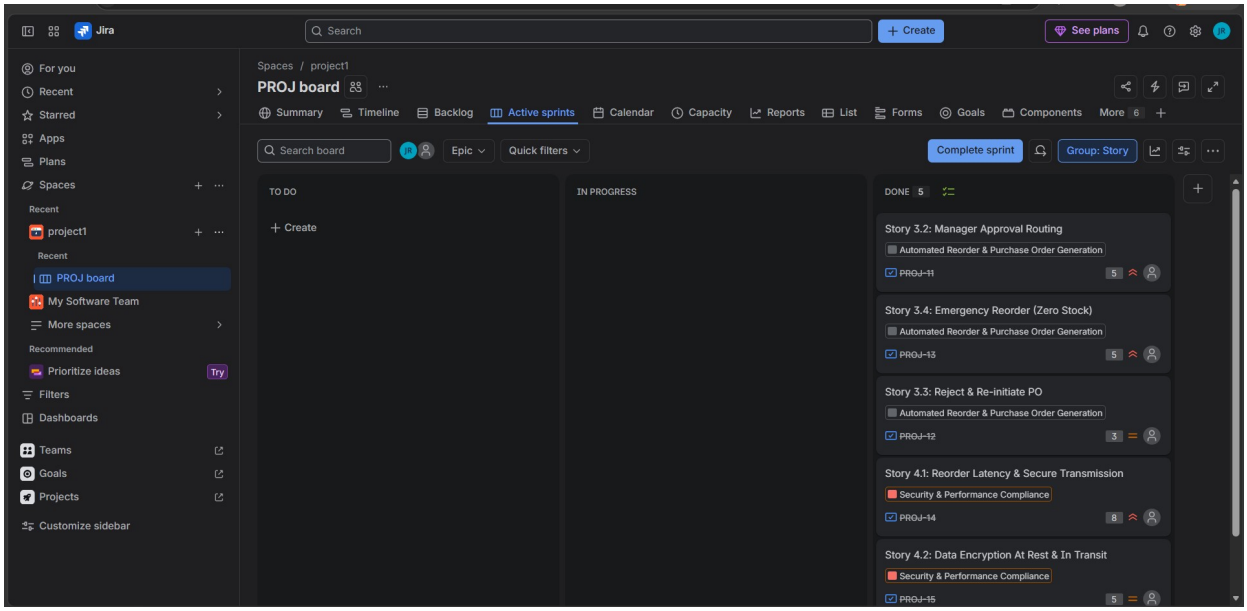
Sprint 1


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Complete sprint
Group: Story
TO DO 2
Story 2.1: Expiry Alert Dashboard
Expiry Monitoring & Alerts
PROJ-8
5
Story 3.1: Auto-Generate Draft PO
Automated Reorder & Purchase Order Generation
PROJ-9
8
Create
IN PROGRESS 2
Story 1.1: Record Batch Receipt
Batch Inventory & FEFO Dispensing
PROJ-6
3
Story 1.2: FEFO Checkout Enforcement
Batch Inventory & FEFO Dispensing
PROJ-7
5
DONE

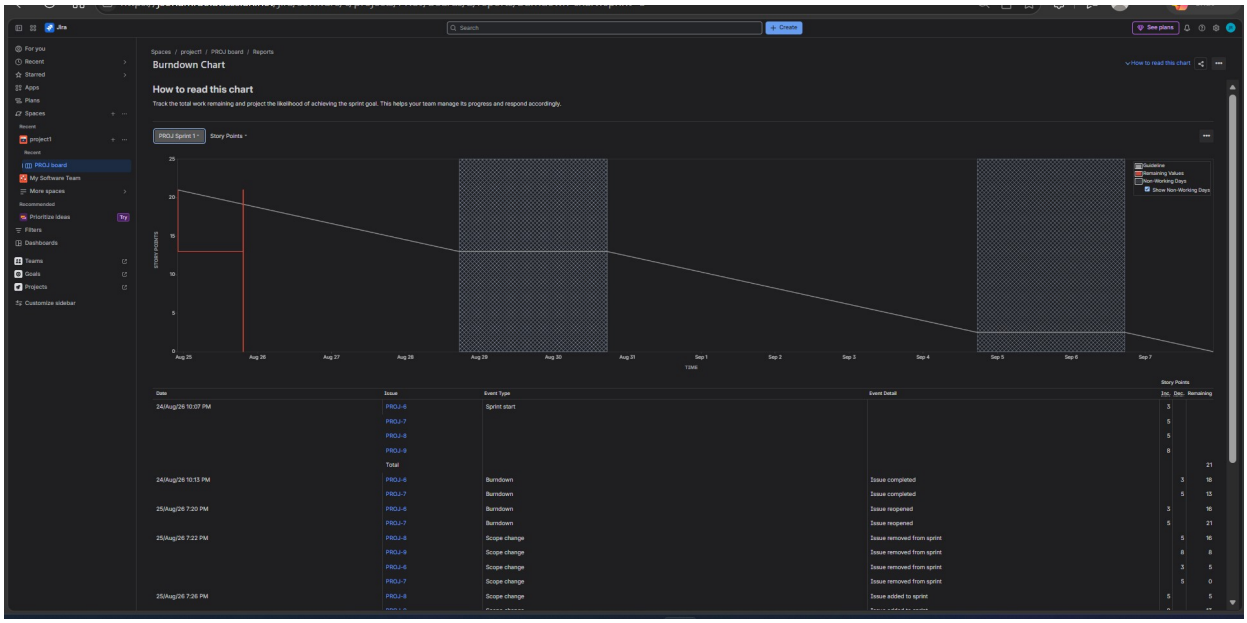

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DONE 4
Story 1.1: Record Batch Receipt
Batch Inventory & FEFO Dispensing
PROJ-6
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Story 1.2: FEFO Checkout Enforcement
Batch Inventory & FEFO Dispensing
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Expiry Monitoring & Alerts
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Story 3.1: Auto-Generate Draft PO
Automated Reorder & Purchase Order Generation
PROJ-9
8
Create

Sprint 2

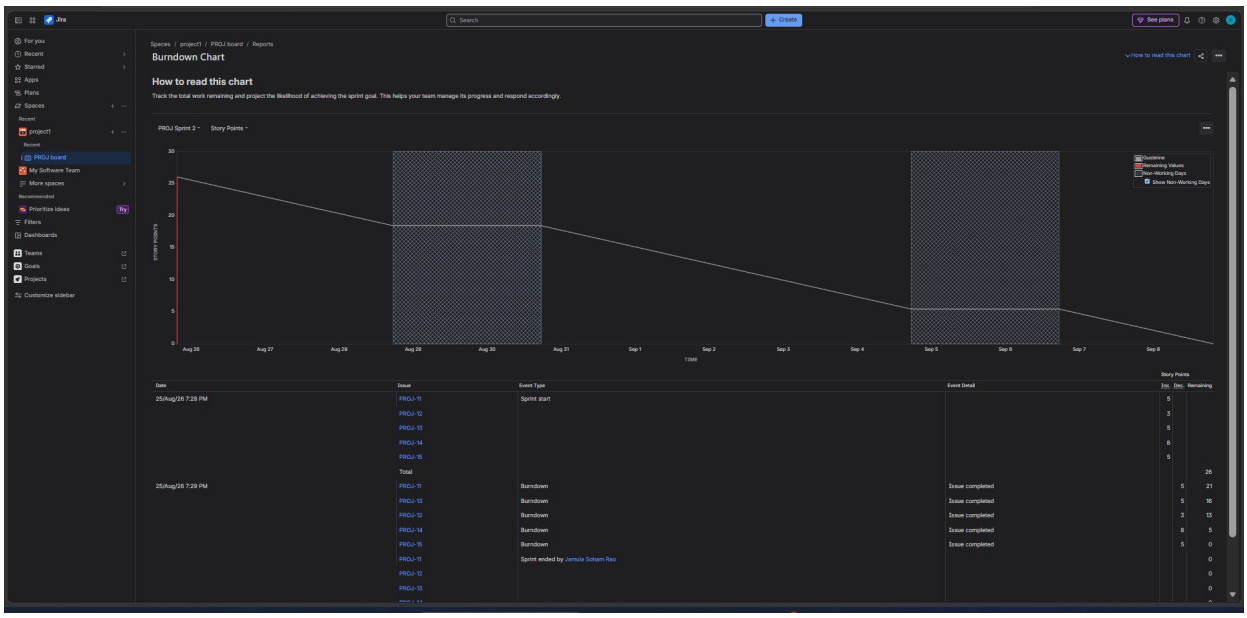

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TO DO 3
IN PROGRESS 2
DONE
Story 3.3: Reject & Re-initiate PO
Automated Reorder & Purchase Order Generation
PROJ-12
3
Story 3.2: Manager Approval Routing
Automated Reorder & Purchase Order Generation
PROJ-11
5
Story 4.1: Reorder Latency & Secure Transmission
Security & Performance Compliance
PROJ-14
8
Story 3.4: Emergency Reorder (Zero Stock)
Automated Reorder & Purchase Order Generation
PROJ-13
5
Story 4.2: Data Encryption At Rest & In Transit
Security & Performance Compliance
PROJ-15
5
Create


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PROJ-11
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Automated Reorder & Purchase Order Generation
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Story 3.3: Reject & Re-initiate PO
Automated Reorder & Purchase Order Generation
PROJ-12
3
Story 4.1: Reorder Latency & Secure Transmission
Security & Performance Compliance
PROJ-14
8
Story 4.2: Data Encryption At Rest & In Transit
Security & Performance Compliance
PROJ-15
5

3. Burndown Charts



Sprint 1 burndown chart.



Sprint 2 burndown chart.

4. Reflection Questions

1. Did your estimations reflect the actual effort?

Mostly yes. Story points were assigned using the Fibonacci scale based on the perceived complexity, dependency load, and uncertainty of each story — for example, Auto-Generate Draft PO (Story 3.1) and Reorder Latency & Secure Transmission (Story 4.1) were both pointed at 8, reflecting that they involve more moving parts (threshold monitoring, supplier lookup, performance/security benchmarks) than a simple form-entry story like Record Batch Receipt (3 points). Since this was a simulated, single-person sprint rather than a real multi-day build, the "effort"

was really the time to model and move each card, so the estimates mainly reflect relative complexity of the underlying requirement rather than measured actual effort.

2. Was your backlog well-prioritized?

Yes, with High/Highest priority given to the requirements that were flagged High in the Lab 1 requirements table (FEFO enforcement, batch recording, PO generation, manager approval, emergency reorder, and the latency/security NFR), and Medium priority reserved for the Lab 1 Medium items (expiry alerting, PO rejection handling, and at-rest encryption). This meant Sprint 1 and Sprint 2 both contained a mix of Highest and Medium priority stories rather than clustering all high-priority work in one sprint, which is reasonable for a 2-sprint simulation but in a longer project would normally front-load the Highest-priority items into Sprint 1.

3. How did your simulated sprint align with your plan?

Sprint 1 diverged from the plan slightly during execution — the burndown log shows PROJ-6 and PROJ-7 being marked complete, then briefly reopened, and PROJ-8/PROJ-9 being removed from and re-added to the sprint scope before final completion, which is visible in the chart as a scope change rather than a clean linear drop. Sprint 2 aligned closely with the plan: all five stories (PROJ-11 through PROJ-15) were completed in sequence without scope changes or reopenings, producing a much cleaner burndown.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 1 chart's scope-change activity suggests that, in a real team setting, mid-sprint reopening and rescoping of stories signals that acceptance criteria weren't fully settled before the sprint started — a sign to spend more time on backlog refinement ahead of sprint planning. Sprint 2's smooth, uninterrupted burndown (26 points cleared with no reopenings) suggests that once the team/process stabilized, the committed scope was well-matched to capacity. Across both sprints, all committed story points were fully burned down by the sprint's actual end rather than left incomplete, indicating the two sprints were not over-committed relative to capacity.